

GeoSSE

Contenido

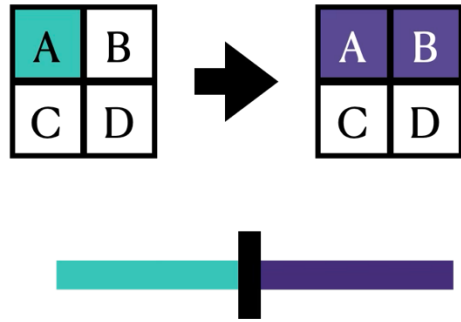
1. Recordatorio DEC
2. Recordatorio ClaSSE
3. GeoSSE
4. Limitaciones GeoSSE
5. Tensorphylo
6. Direcciones futuras en Biogeo

1) Recordatorio DEC

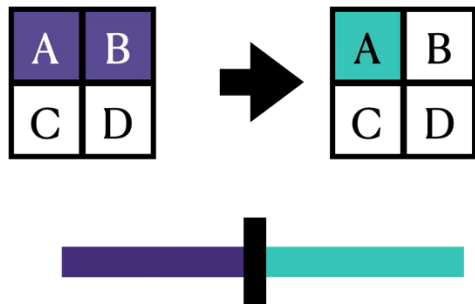
Anatomy of a DEC model

Anagenetic events

Dispersal

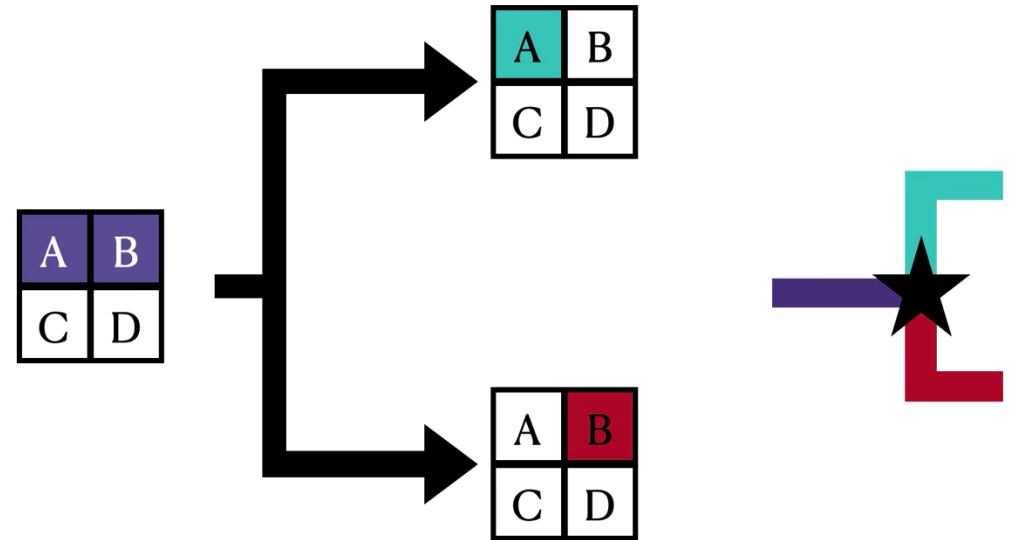


Extinction



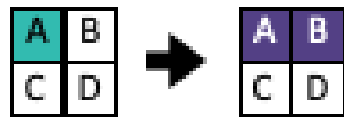
Cladogenetic events

Associated to cladogenesis

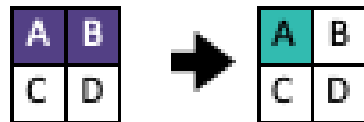


Modelo DEC – Eventos anagenéticos

Dispersal



Extinction



$Q =$

	$\emptyset$	A	B	C	AB	AC	BC	ABC
$\emptyset$	-	0	0	0	0	0	0	0
A	e	-	0	0	d	d	0	0
B	e	0	-	0	d	0	d	0
C	e	0	0	-	0	d	d	0
AB	0	e	e	0	-	0	0	2d
AC	0	e	0	e	0	-	0	2d
BC	0	0	e	e	0	0	-	2d
ABC	0	0	0	0	e	e	e	-

¿Qué significa modelar un proceso como eventos estocásticos?

- Parametrizamos los eventos de acuerdo a las tasas (rates) a las que ocurren, e.g., tasa de especiación, tasa de extinción, etc.

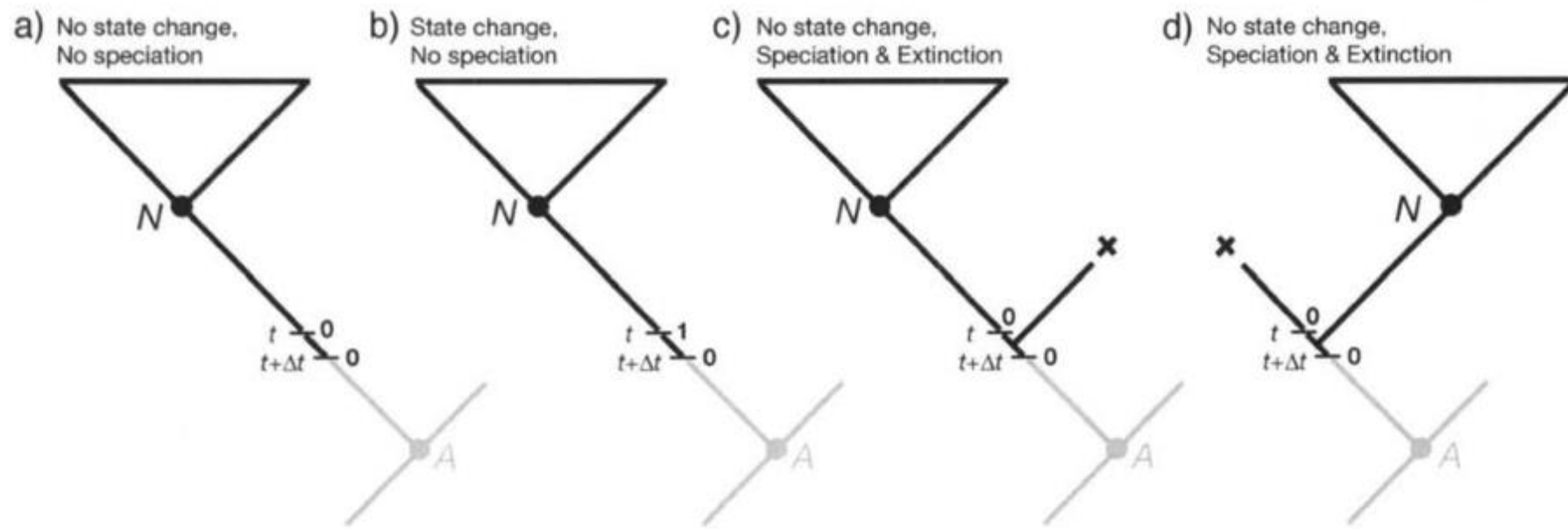


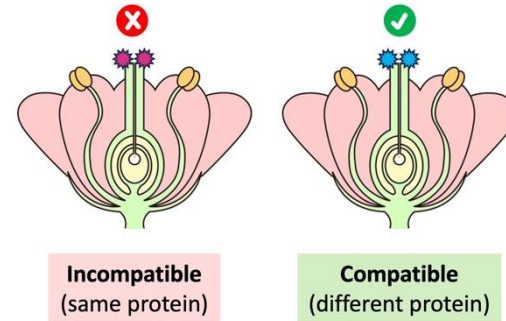
FIGURE 2. Alternative scenarios by which a lineage with state 0 at time $t + \Delta t$ on the branch might yield clade descended from node N but no other living descendants.

2) Recordatorio ClaSSE

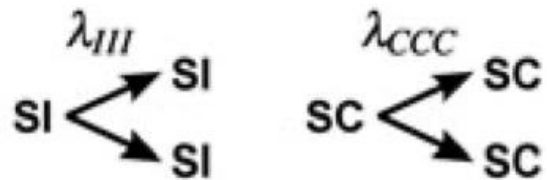
ClaSSE

(Cladogenetic events)

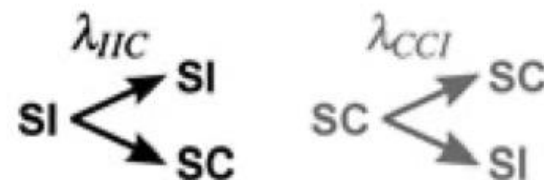
Evolución de la compatibilidad sexual en Solanaceae.



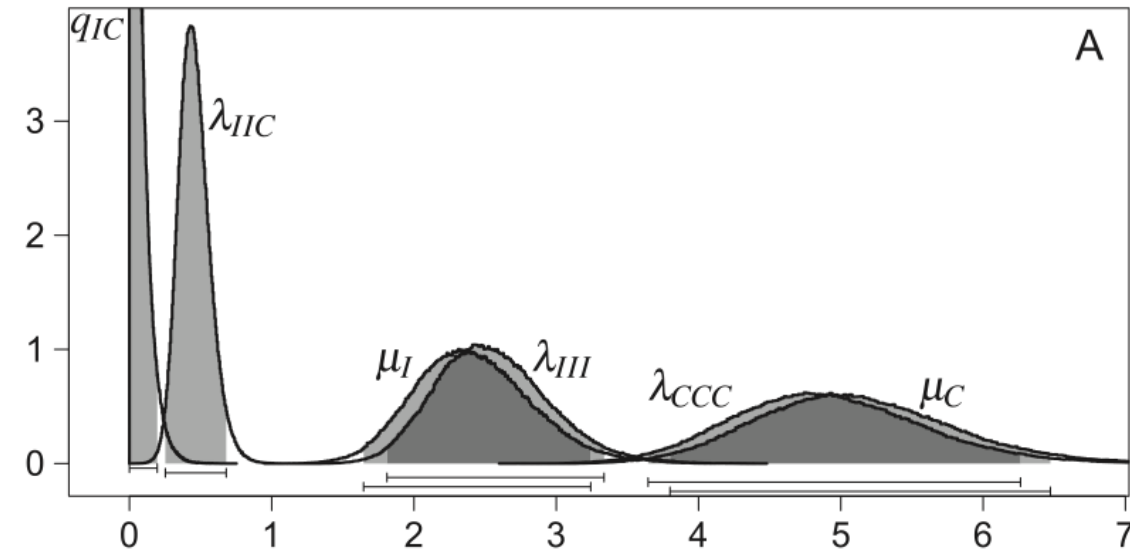
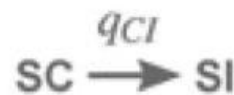
Cladogenesis, no state change
(BiSSE & ClaSSE)



Cladogenetic state change
(ClaSSE only)



Anagenetic state change (BiSSE & ClaSSE)



3) GeoSSE

Phylogenetic Inference of Reciprocal Effects between Geographic Range Evolution and Diversification

Emma E. Goldberg , Lesley T. Lancaster, Richard H. Ree [Author Notes](#)

Systematic Biology, Volume 60, Issue 4, July 2011, Pages 451–465, <https://doi.org/10.1093/sysbio/syr046>

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Abstract

Geographic characters—traits describing the spatial distribution of a species—may both affect and be affected by processes associated with lineage birth and death. This is potentially confounding to comparative analyses of species distributions because current models do not allow reciprocal interactions between the evolution of ranges and the growth of phylogenetic trees. Here, we introduce a

The GeoSSE model

Notación original de Goldberg et al. (2011)

Our model of “geographic state speciation and extinction,” GeoSSE, differs from the “binary state speciation and extinction” model (BiSSE, [Maddison et al. 2007](#)) in three specific ways. First, the character states are defined so that a single, wide-ranging species may occupy more than one area or region simultaneously. Second, the transitions between states are explicitly parameterized in terms of range expansion through dispersal and range contraction through local extirpation. And third, rates of speciation, extinction, and dispersal are tied to the regions themselves rather than to the character states, causing the dynamics of wider-ranging species to be determined by the cumulative effects of the areas they inhabit.

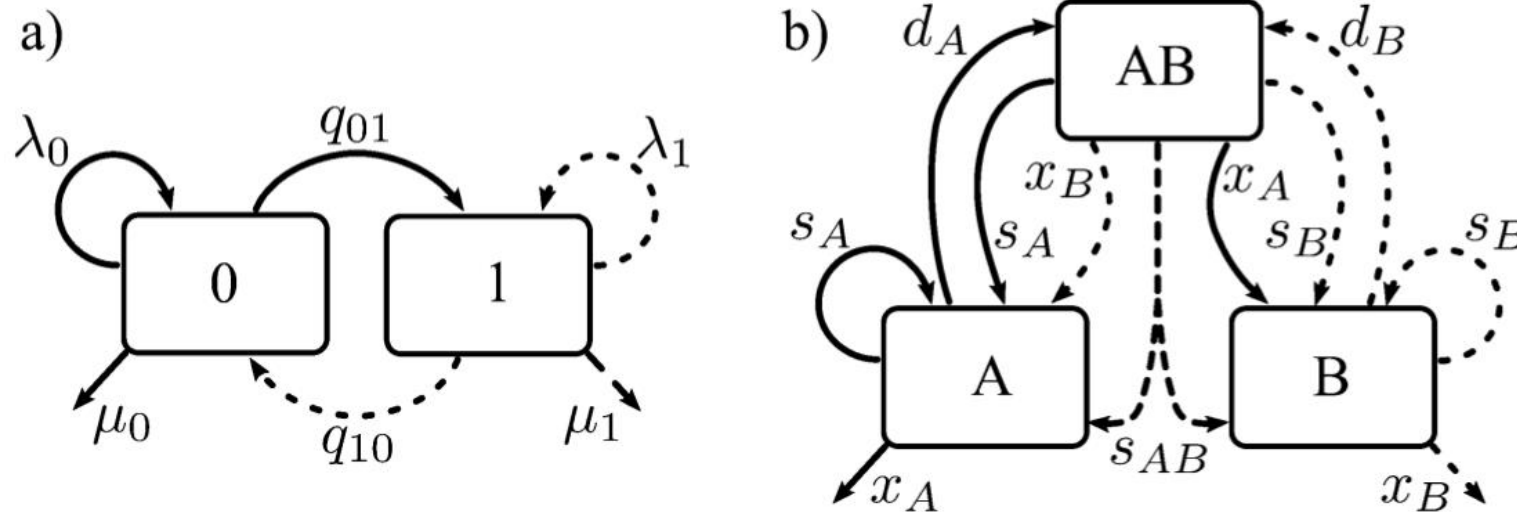
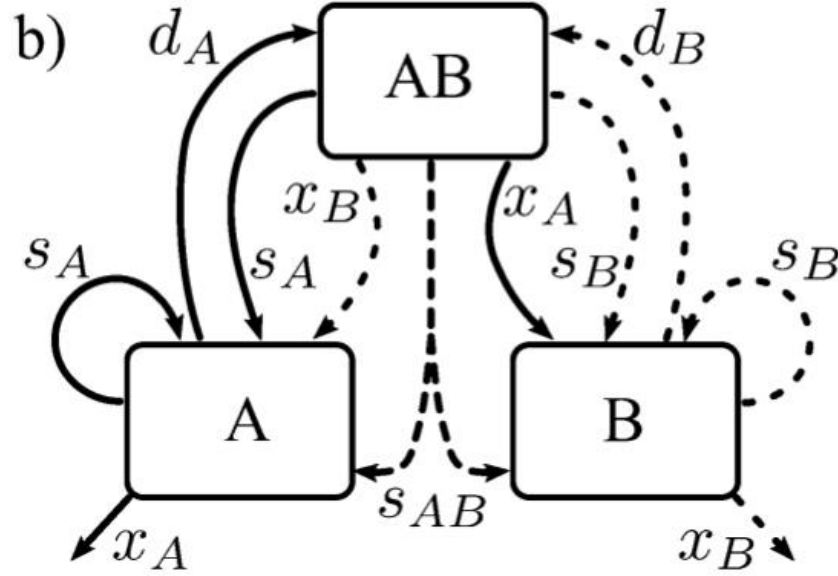


FIGURE 1. The states and allowed transitions in a) BiSSE and b) GeoSSE. Both models have six rate parameters and allow state-dependent speciation (λ or s) and extinction (μ or x), and anagenetic state changes (q , d , x). GeoSSE additionally allows state changes during speciation in association with the third (AB) character state.



$$\frac{dD_{NA}}{dt} = -(s_A + d_A + x_A)D_{NA}(t) + d_A D_{NAB}(t) + 2s_A D_{NA}(t)E_A(t),$$

$$\frac{dD_{NB}}{dt} = -(s_B + d_B + x_B)D_{NB}(t) + d_B D_{NAB}(t) + 2s_B D_{NB}(t)E_B(t),$$

$$\begin{aligned} \frac{dD_{NAB}}{dt} = & -(s_A + s_B + s_{AB} + x_A + x_B)D_{NAB}(t) \\ & + x_A D_{NB}(t) + x_B D_{NA}(t) \\ & + s_A [E_A(t)D_{NAB}(t) + E_{AB}(t)D_{NA}(t)] \\ & + s_B [E_B(t)D_{NAB}(t) + E_{AB}(t)D_{NB}(t)] \\ & + s_{AB} [E_A(t)D_{NB}(t) + E_B(t)D_{NA}(t)], \end{aligned}$$

FIGURE 1. The states and allowed transitions in a) BiSSE and b) GeoSSE. Both models have six rate parameters and allow state-dependent speciation (λ or s) and extinction (μ or x), and anagenetic state changes (q , d , x). GeoSSE additionally allows state changes during speciation in association with the third (AB) character state.

Verosimilitud de eventos cladogenéticos

$$D_{CA}(t_C) = D_{NA}(t_C)D_{MA}(t_C)s_A, \quad (2a)$$

$$D_{CB}(t_C) = D_{NB}(t_C)D_{MB}(t_C)s_B, \quad (2b)$$

$$\begin{aligned} D_{CAB}(t_C) = & \frac{1}{2}[D_{NAB}(t_C)D_{MA}(t_C) + D_{NA}(t_C)D_{MAB}(t_C)]s_A \\ & + \frac{1}{2}[D_{NAB}(t_C)D_{MB}(t_C) + D_{NB}(t_C)D_{MAB}(t_C)]s_B \\ & + \frac{1}{2}[D_{NA}(t_C)D_{MB}(t_C) + D_{NB}(t_C)D_{MA}(t_C)]s_{AB}. \end{aligned} \quad (2c)$$

Verosimilitud de extinción

$$\frac{dE_A}{dt} = -(s_A + d_A + x_A)E_A(t) + x_A + d_A E_{AB}(t) + s_A E_A(t)^2, \quad (3a)$$

$$\frac{dE_B}{dt} = -(s_B + d_B + x_B)E_B(t) + x_B + d_B E_{AB}(t) + s_B E_B(t)^2, \quad (3b)$$

$$\begin{aligned} \frac{dE_{AB}}{dt} = & -(s_A + s_B + s_{AB} + x_A + x_B)E_{AB}(t) + x_A E_B(t) \\ & + x_B E_A(t) + s_A E_{AB}(t)E_A(t) + s_B E_{AB}(t)E_B(t) \\ & + s_{AB} E_A(t)E_B(t). \end{aligned} \quad (3c)$$



Chaparral vs forest

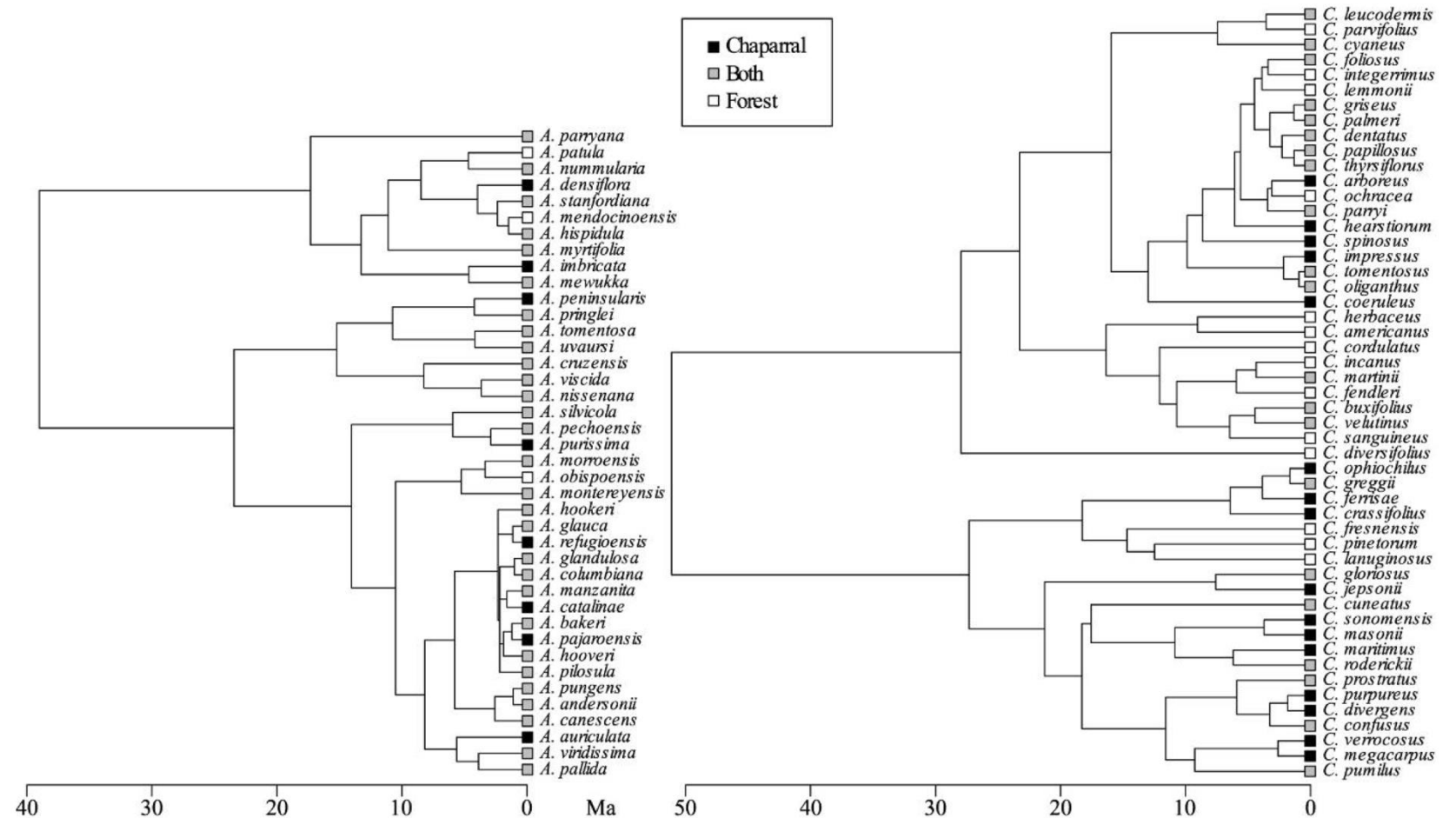


FIGURE 6. Summary phylogenies with tip states for *Arctostaphylos* and *Ceanothus*. These are the maximum clade credibility trees with mean node heights; the posterior set of trees was used in the analysis.

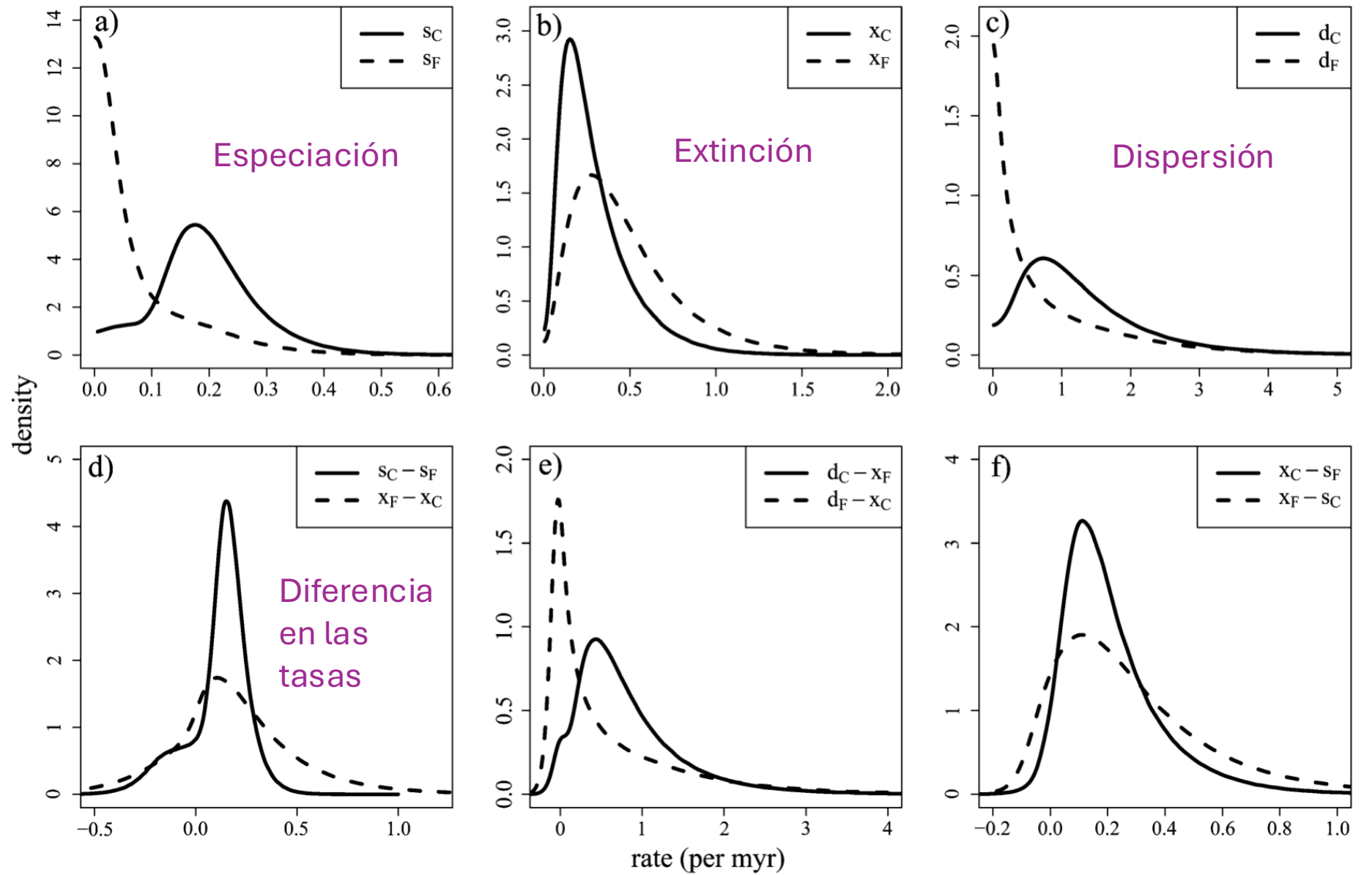
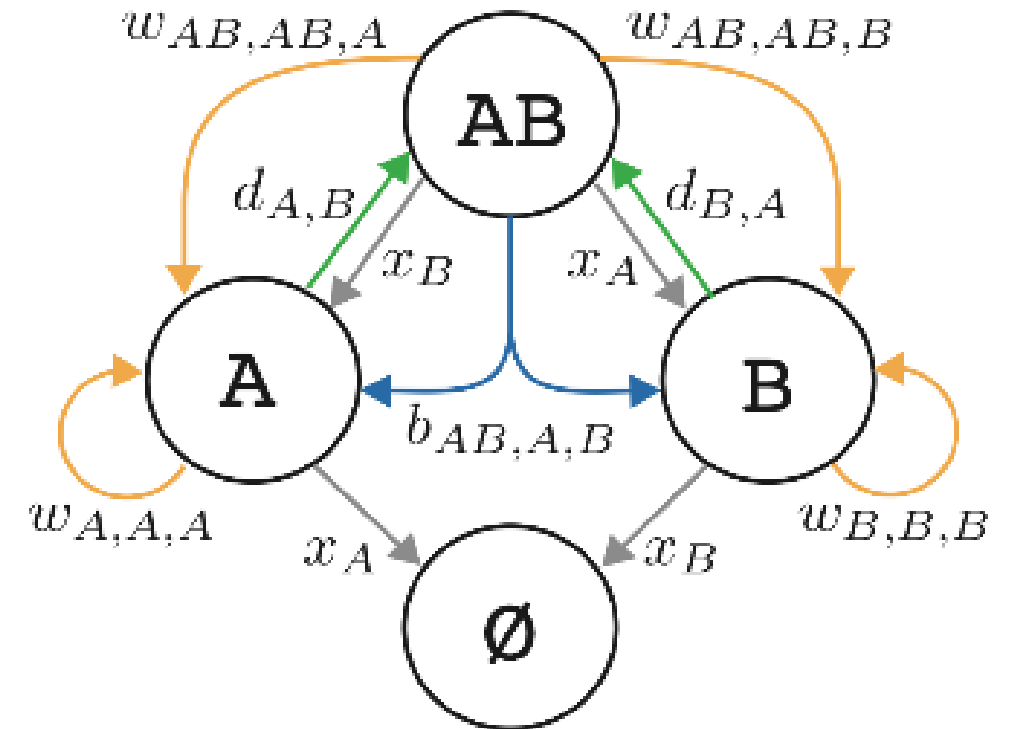
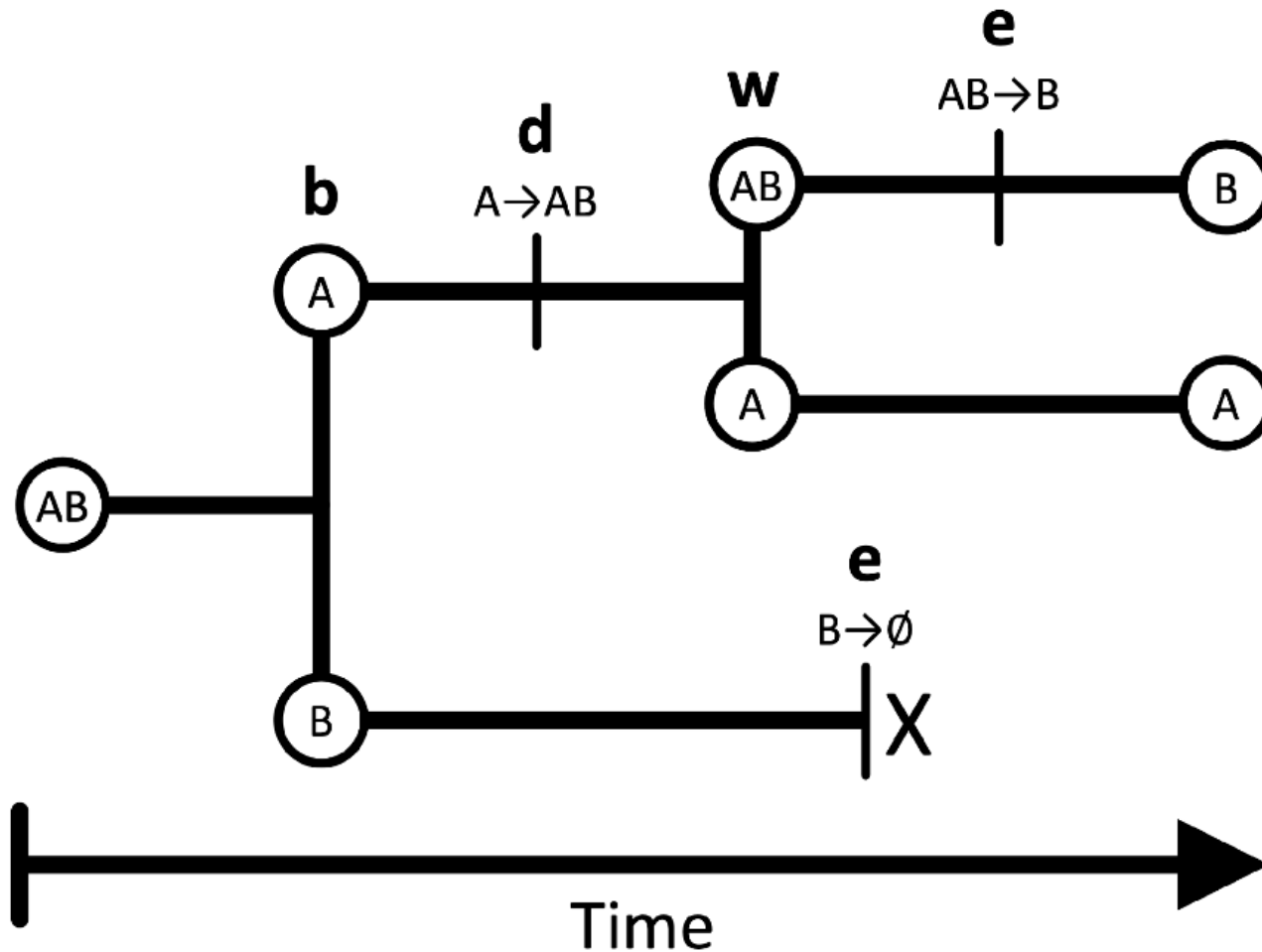


FIGURE 7. Posterior probability distributions of **GeoSSE** parameters for California plants. Rates of a) speciation, b) extinction, and c) dispersal were estimated for chaparral (solid lines) and forest (dashed lines) habitats. Differences in d) speciation and extinction rates between habitats, e) range contraction versus expansion, and f) mechanism of specialization were computed from the six estimated parameters.

Generalizando GeoSSE

Notación utilizada en RevBayes (Landis et al)



Anagenetic events

(d)ispersal
e(x)tinction

Cladogenetic events

(w)ithin-region
speciation
(b)etween-region
speciation

¿Pero y si nos interesan más de dos regiones?

- El numero de tasas aumenta linealmente con el número de regiones.
- Pero el número de combinaciones de rangos y listado de eventos posibles aumenta exponencialmente

$Q =$

	$\emptyset$	A	B	C	AB	AC	BC	ABC
$\emptyset$	—	0	0	0	0	0	0	0
A	e_A	—	0	0	d_{AB}	d_{AC}	0	0
B	e_B	0	—	0	d_{BA}	0	d_{BC}	0
C	e_C	0	0	—	0	d_{CA}	d_{CB}	0
AB	0	e_B	e_A	0	—	0	0	$d_{AC} + d_{BC}$
AC	0	e_C	0	e_A	0	—	0	$d_{AB} + d_{CB}$
BC	0	0	e_C	e_B	0	0	—	$d_{BA} + d_{CA}$
ABC	0	0	0	0	e_C	e_B	e_A	—

$$2^4 = 16$$

$$2^5 = 32$$

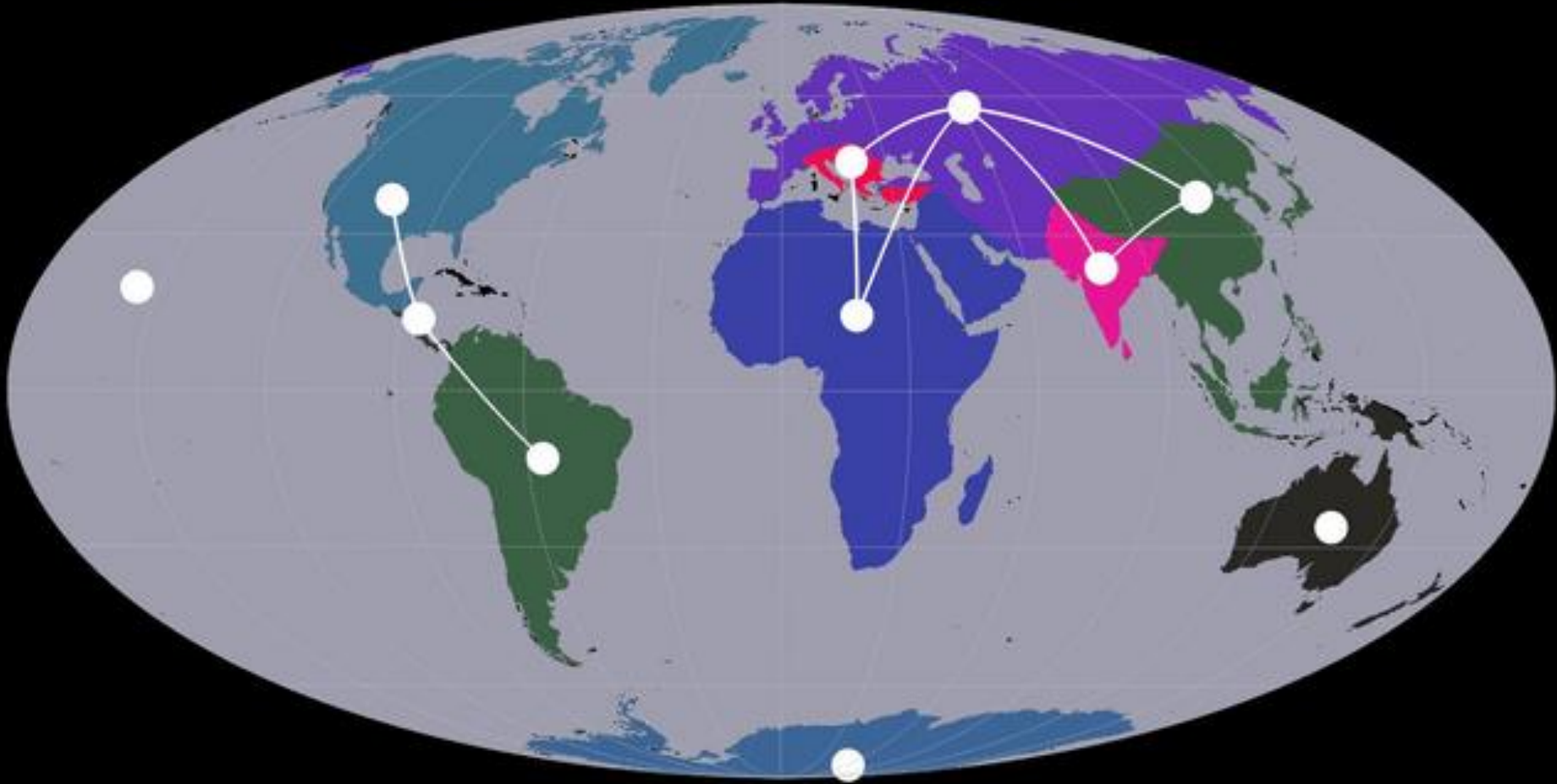
$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

The **paleogeography** information



The paleogeographically informed GeoSSE +J model

Anagenetic events



adjacent
dispersal



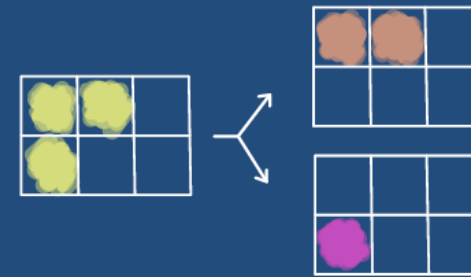
long-distance
dispersal



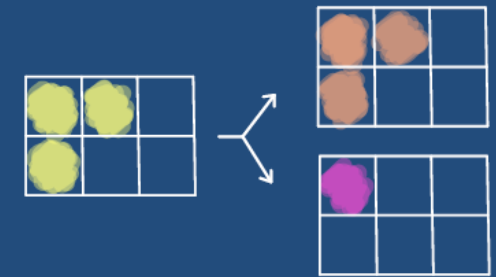
extirpation



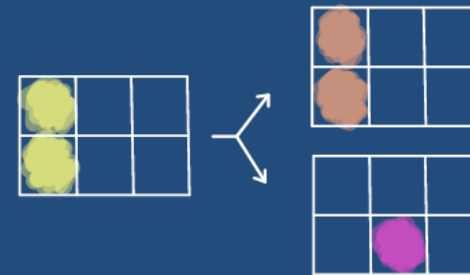
Cladogenetic events



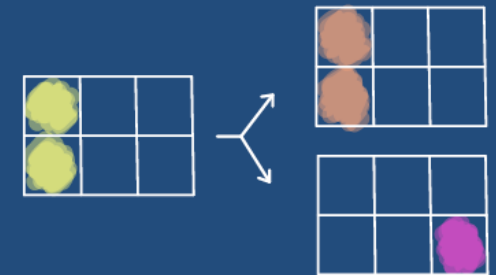
allopatric speciation



sympatric speciation



adjacent jump

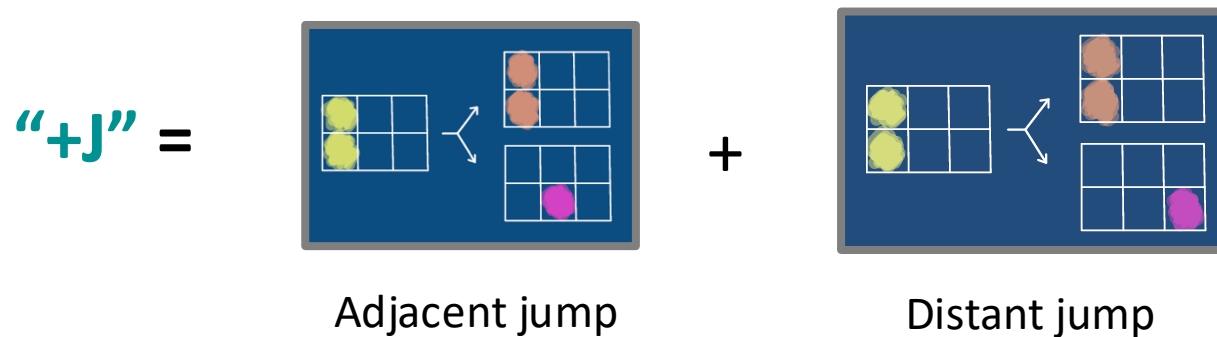
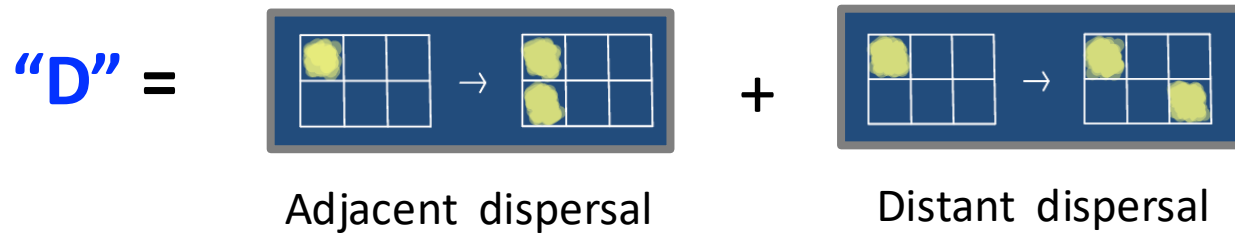


long-distance jump

Results

Parameter estimates

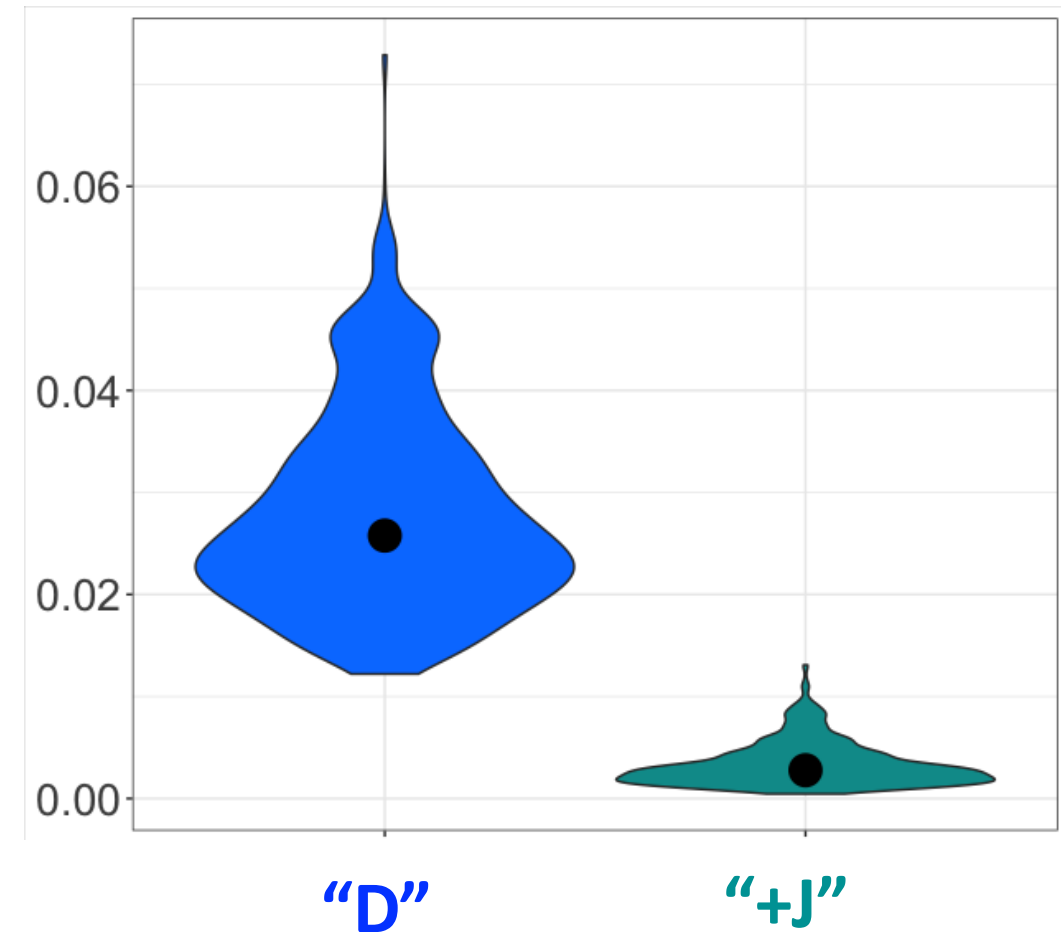
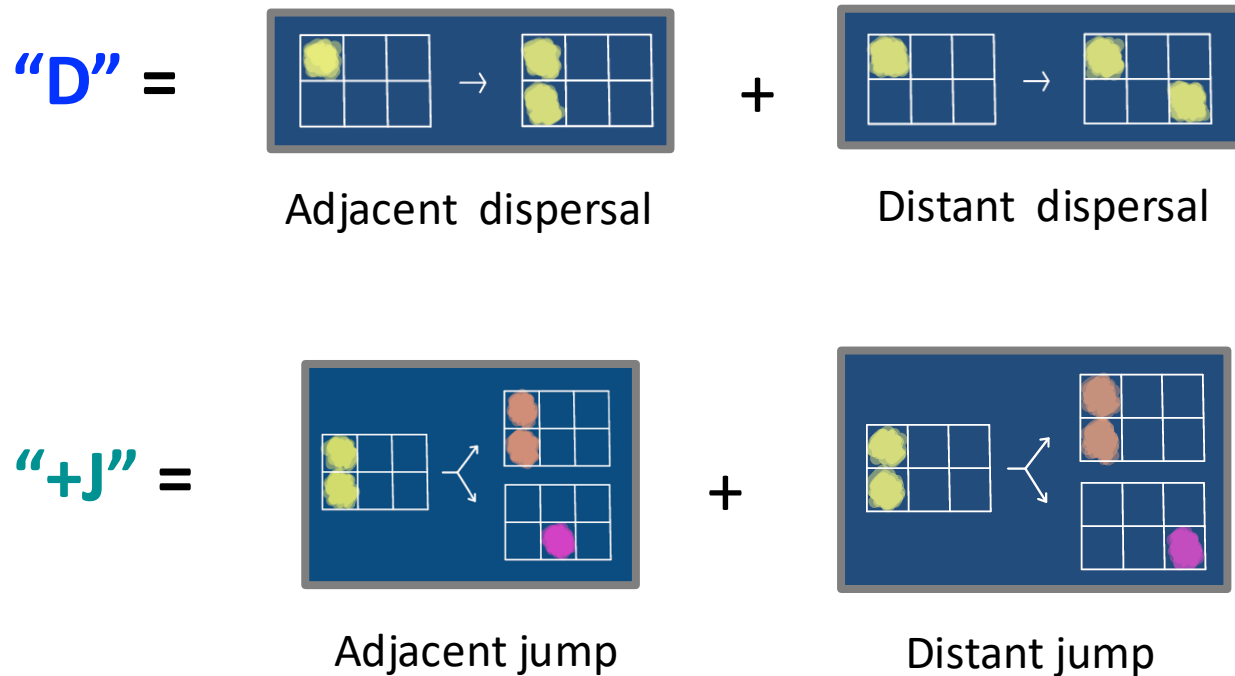
Does the GeoSSE framework fix the overestimation of the jump dispersal (+J) parameter?



Results

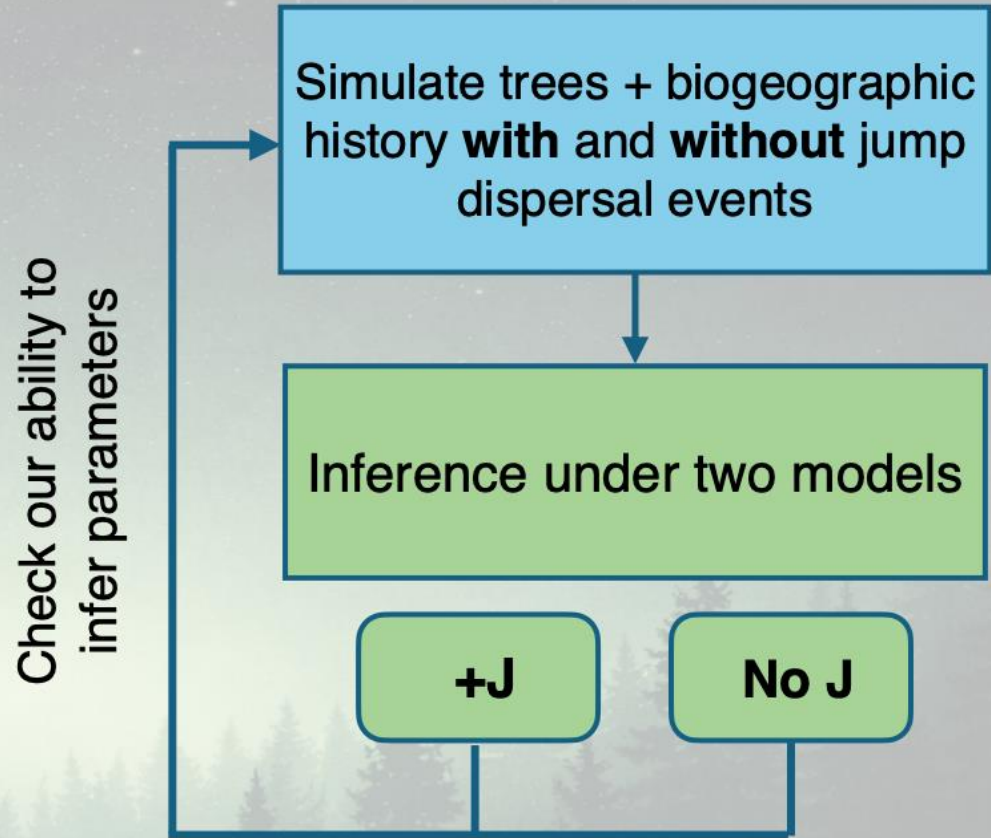
Parameter estimates

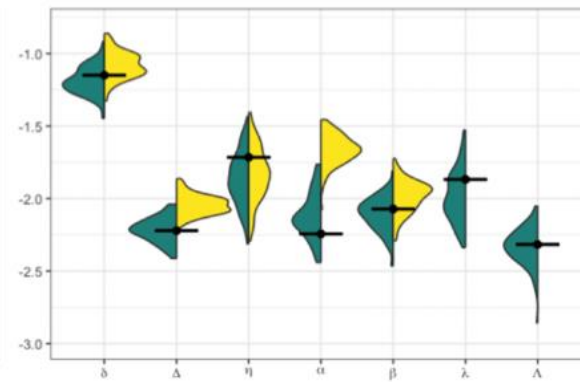
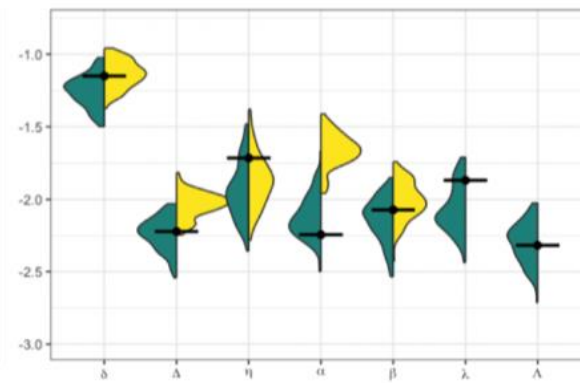
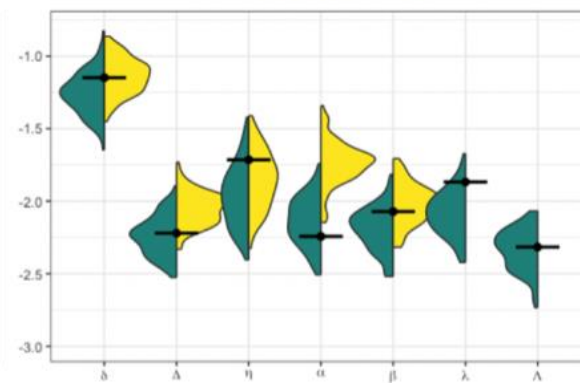
Does the GeoSSE framework fix the overestimation of the jump dispersal (+J) parameter?



Can we accurately
infer biogeographic
parameters using
our model?

Simulations





The J parameter

		Simulated data	
Inference model	No J	No jump events  True model	With jump events Underparametrization Considerably biased estimates of subset speciation, and distant dispersal.
	+J	Overparametrization Slightly biased non-jump parameters. Not-zero values of jump parameters.	 True model

4) Limitaciones GeoSSE

Ventajas y limitantes de GeoSSE

GeoSSE is a software package for simulating the spread of invasive species. It is designed to be user-friendly and flexible, allowing users to simulate the spread of a species from a single point source or multiple sources. The software can be used to simulate the spread of a species over a large geographic area, and it can be used to evaluate the impact of different management scenarios. GeoSSE is a powerful tool for understanding the spread of invasive species and for developing effective management strategies.

5) Tensorphylo

Diapos cortesía de Mike May

Implementations

[illegible]

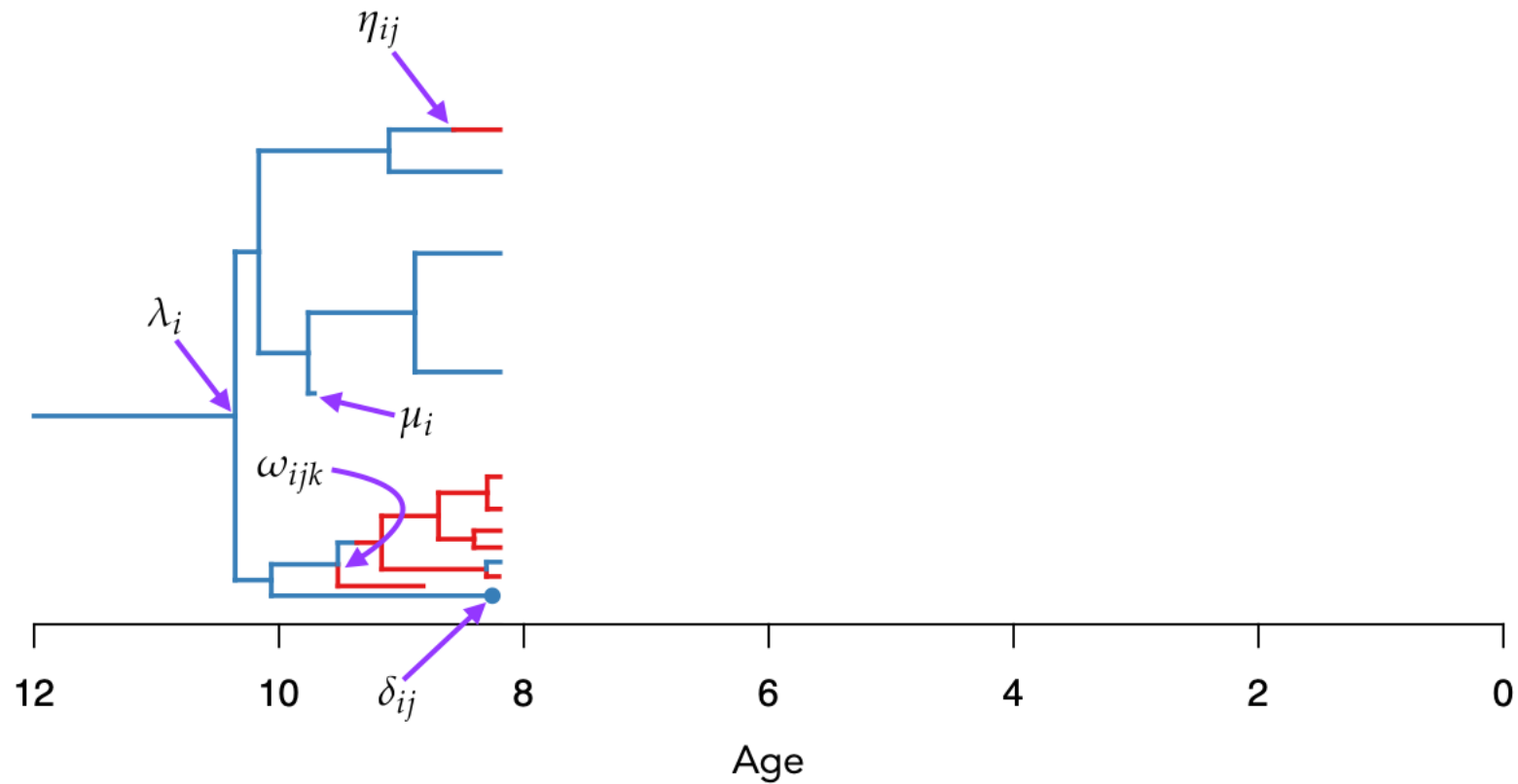
A general state-dependent birth-death model

Lineage-specific events		Tree-wide events	
Speciation:	$\lambda_i(t)$	$v_i(t)$	
Extinction:	$\mu_i(t)$	$\gamma_i(t)$	
Sampling:	$\phi_i(t)$	$\rho_i(t)$	
Treatment:	$\delta_i(t)$	$\xi_i(t)$	
State-change events			
		Anagenetic:	$\eta_{ij}(t)$
		Cladogenetic:	$\omega_{ijk}(t)$
		Mass-anagenesis:	$\zeta_{ij}(t)$

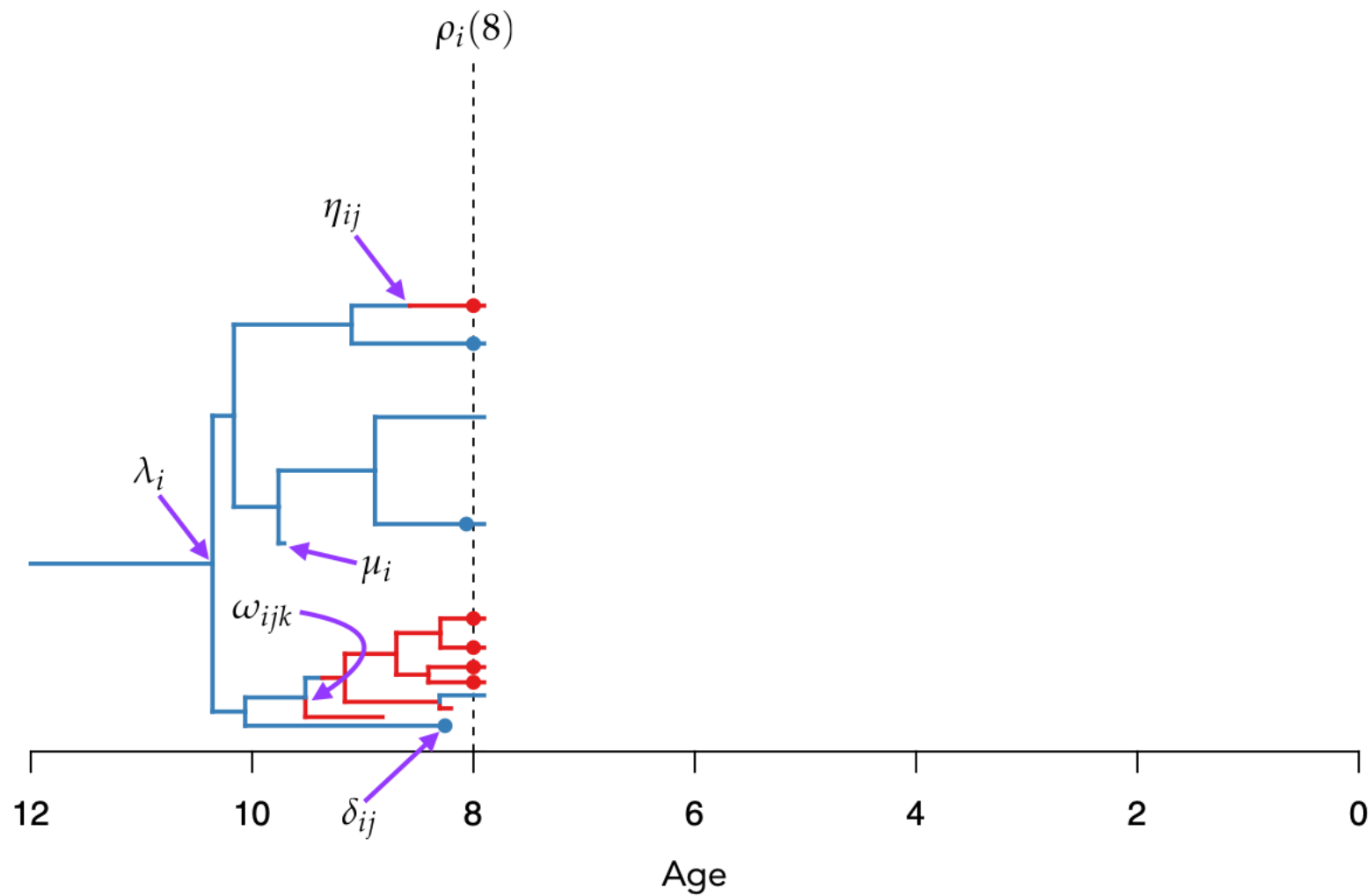
A general state-dependent birth-death model



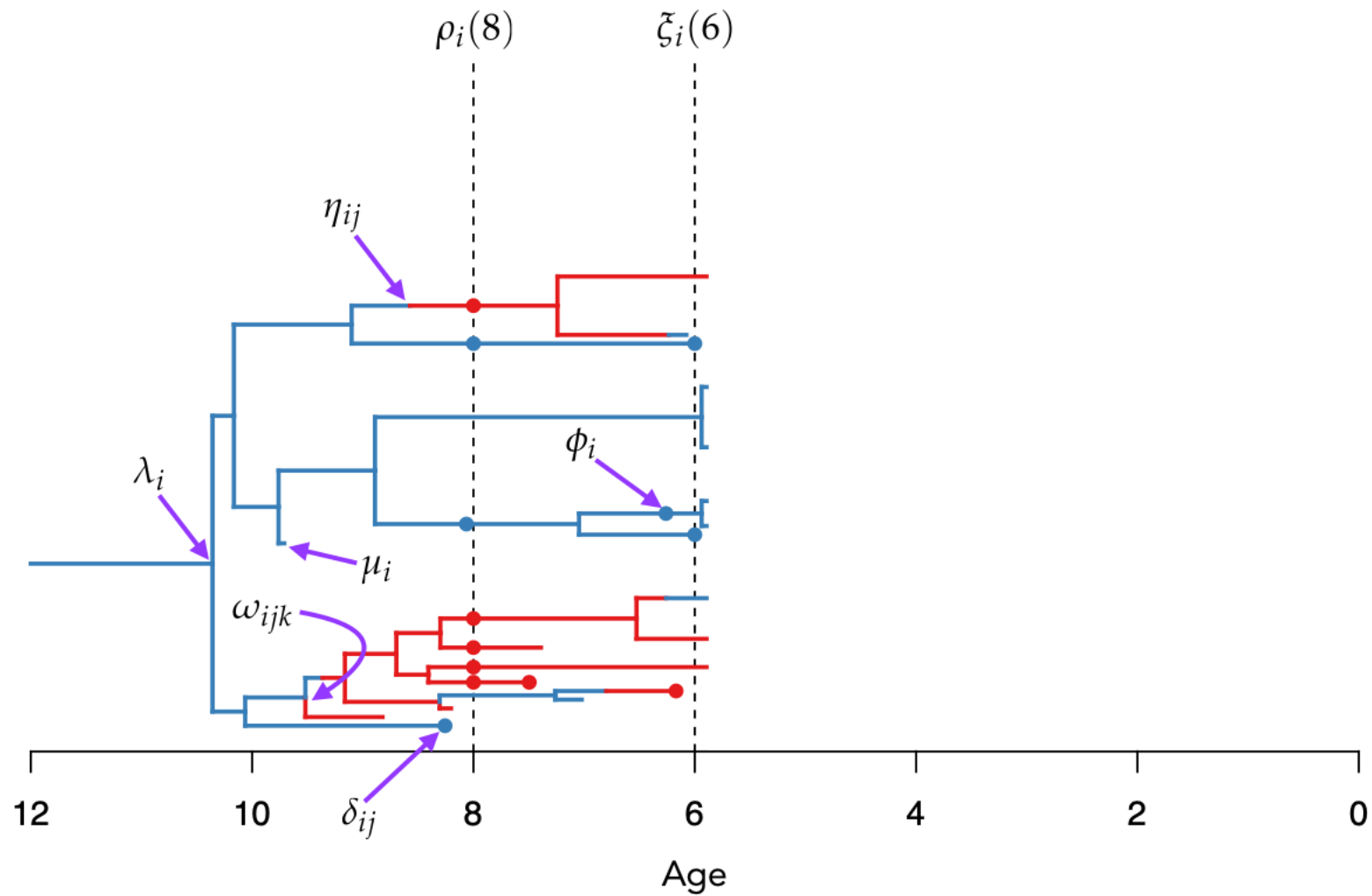
A general state-dependent birth-death model



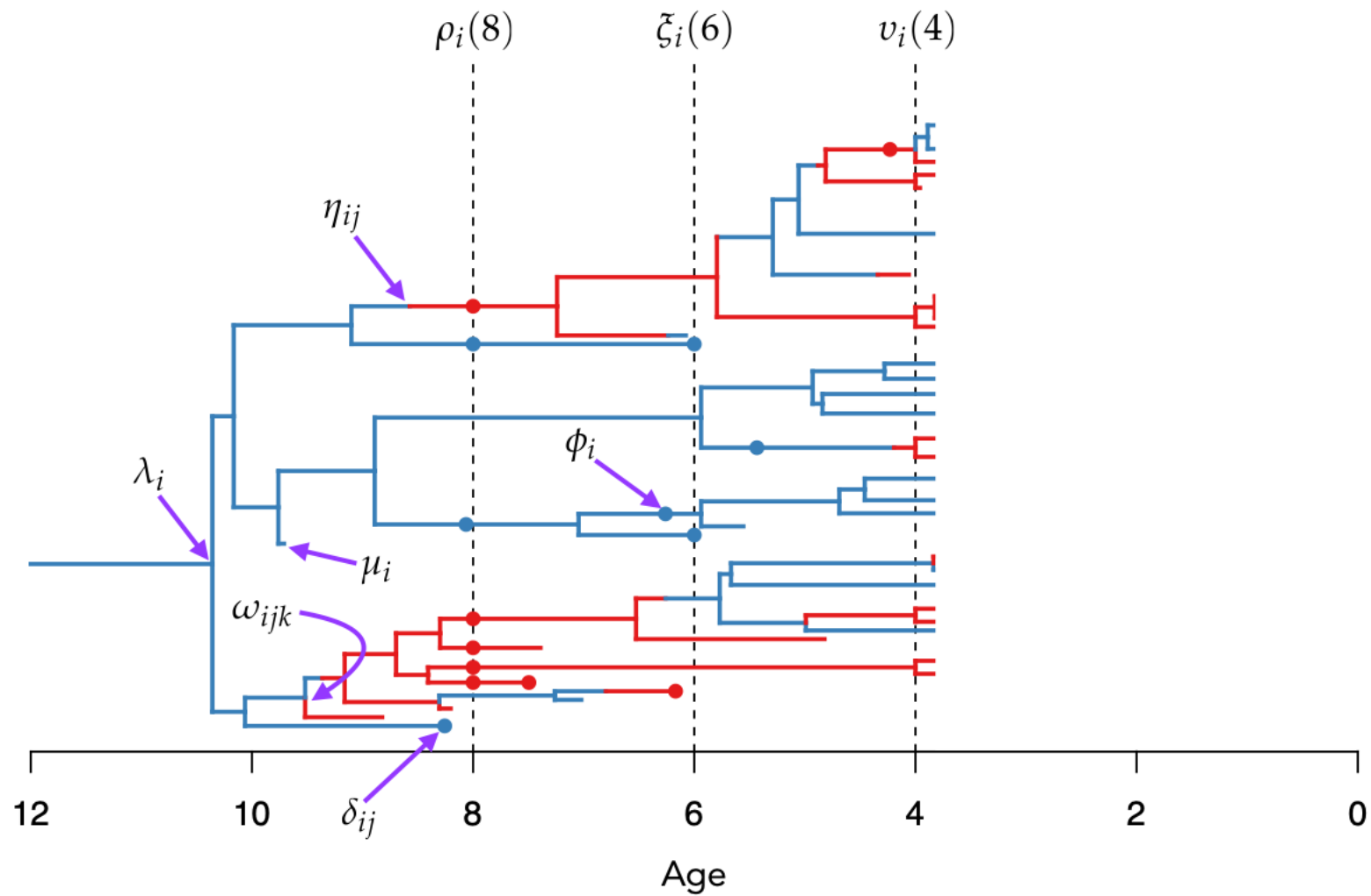
A general state-dependent birth-death model



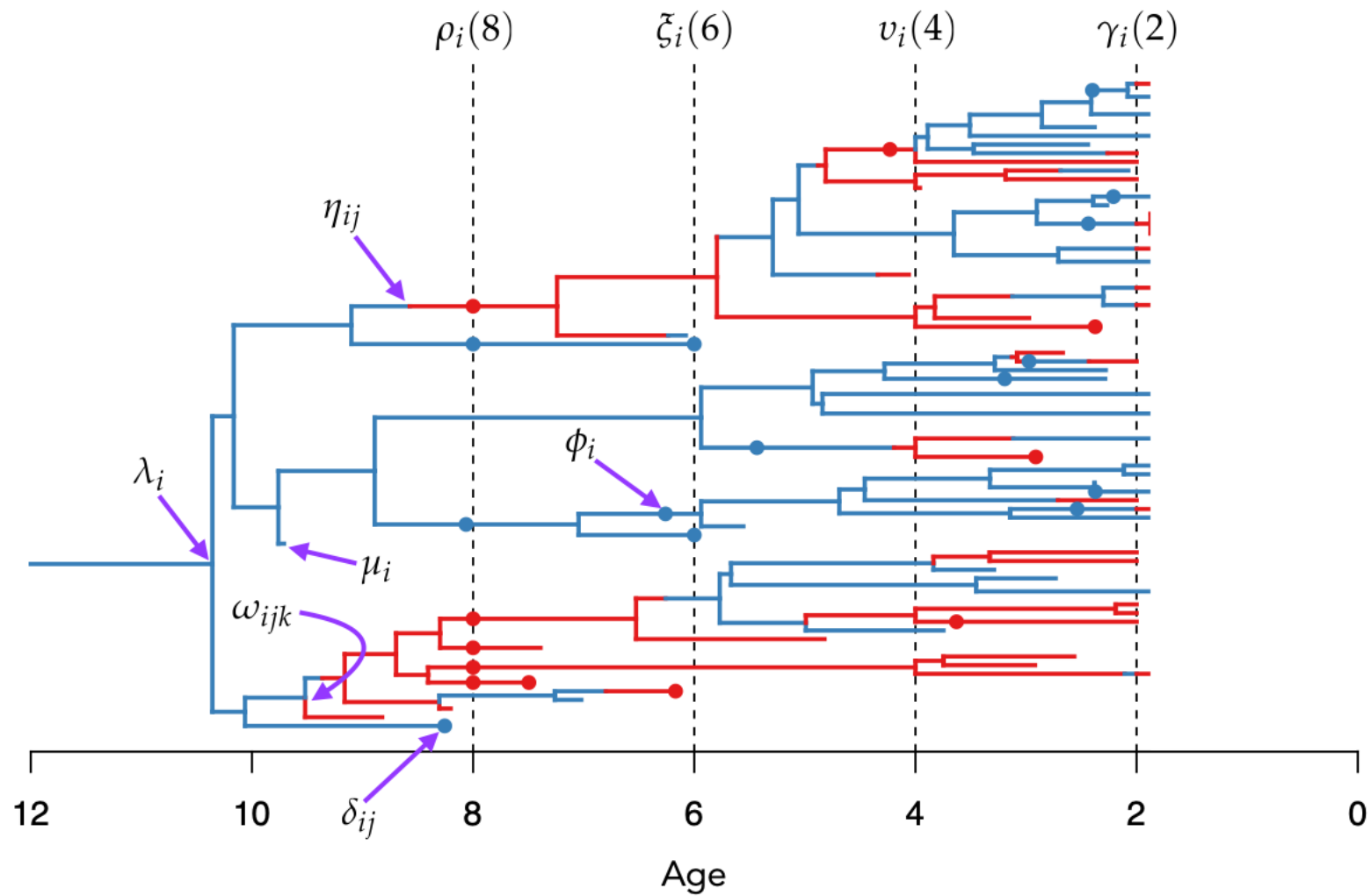
A general state-dependent birth-death model



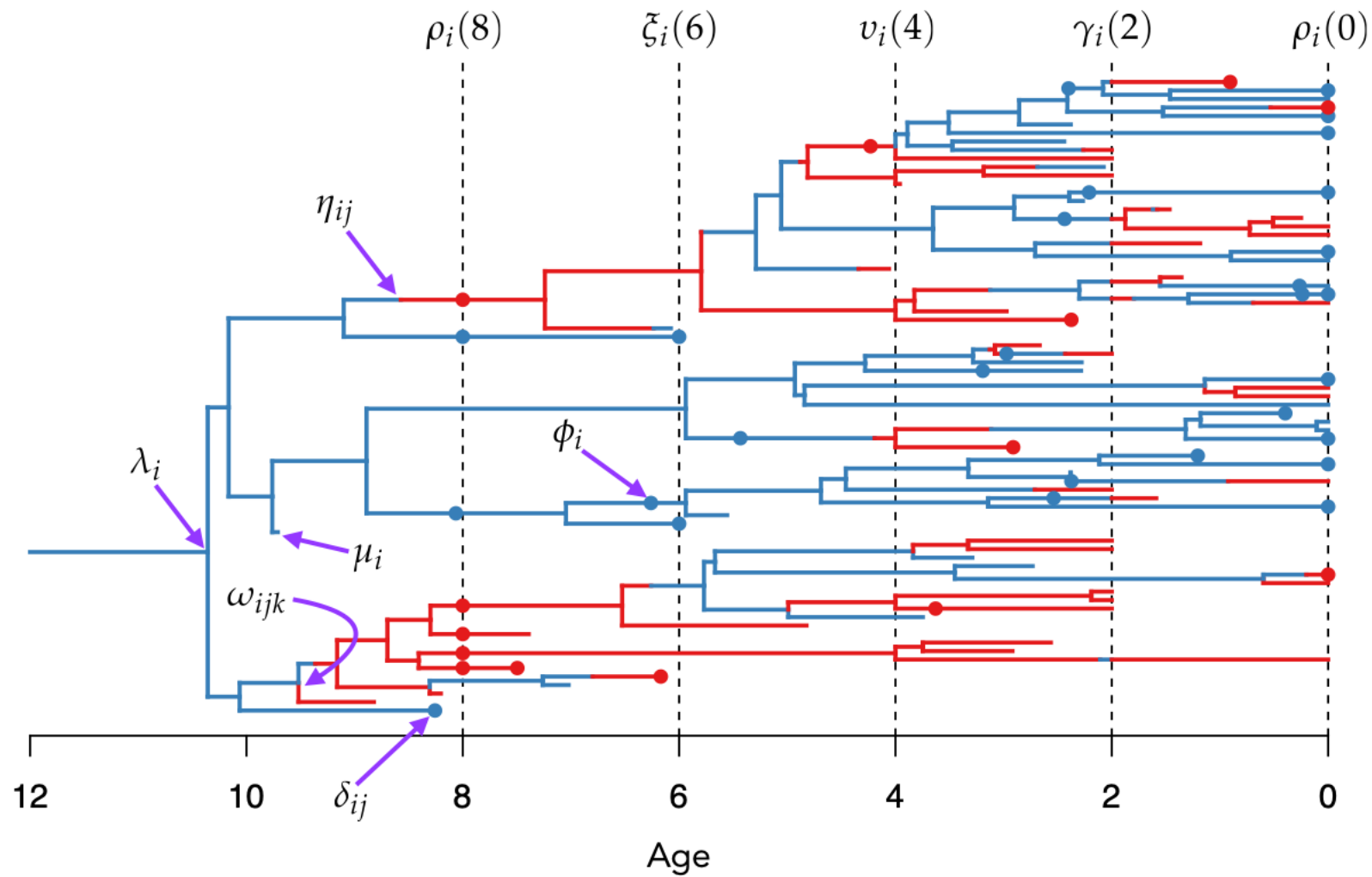
A general state-dependent birth-death model



A general state-dependent birth-death model



A general state-dependent birth-death model



TensorPhylo

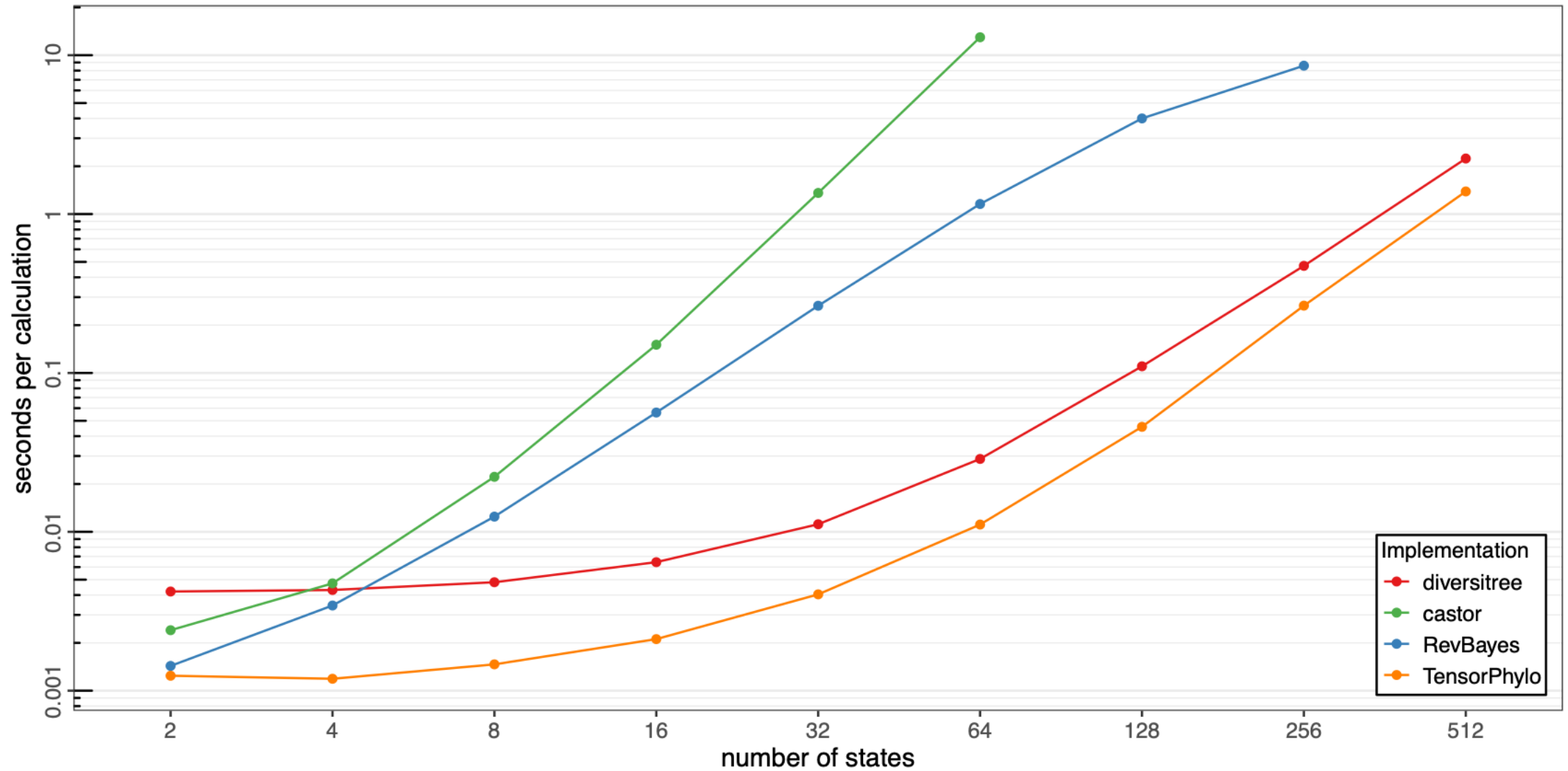
TensorPhylo is a c++ library that computes probabilities of trees under the general state-dependent model (*it doesn't do inference*).

We use the linear algebra library **Eigen** for all our matrix and tensor operations.

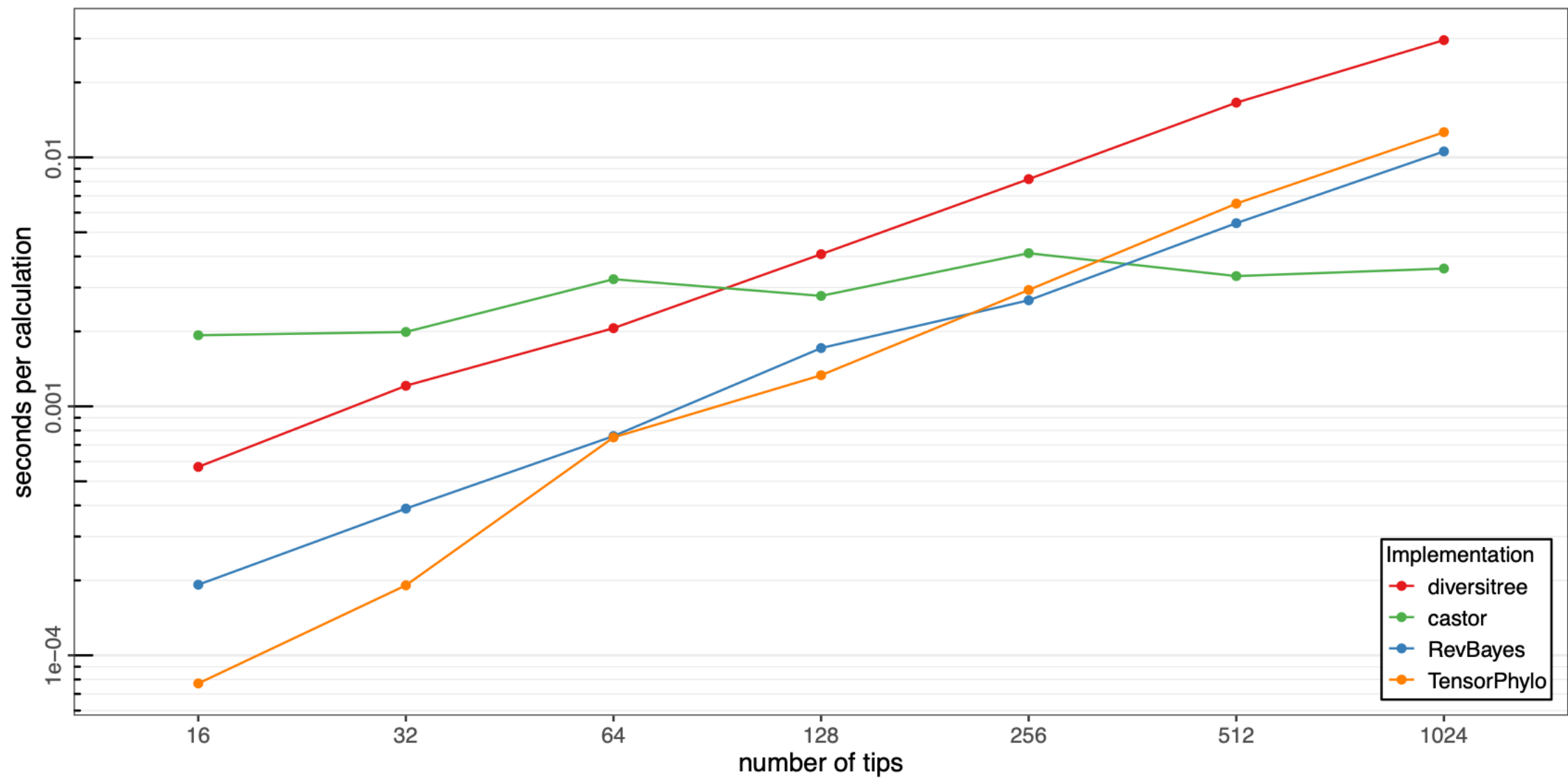
We use adaptive numerical integrators from **Boost odeint**.

We use **OpenMP** for parallel processing.

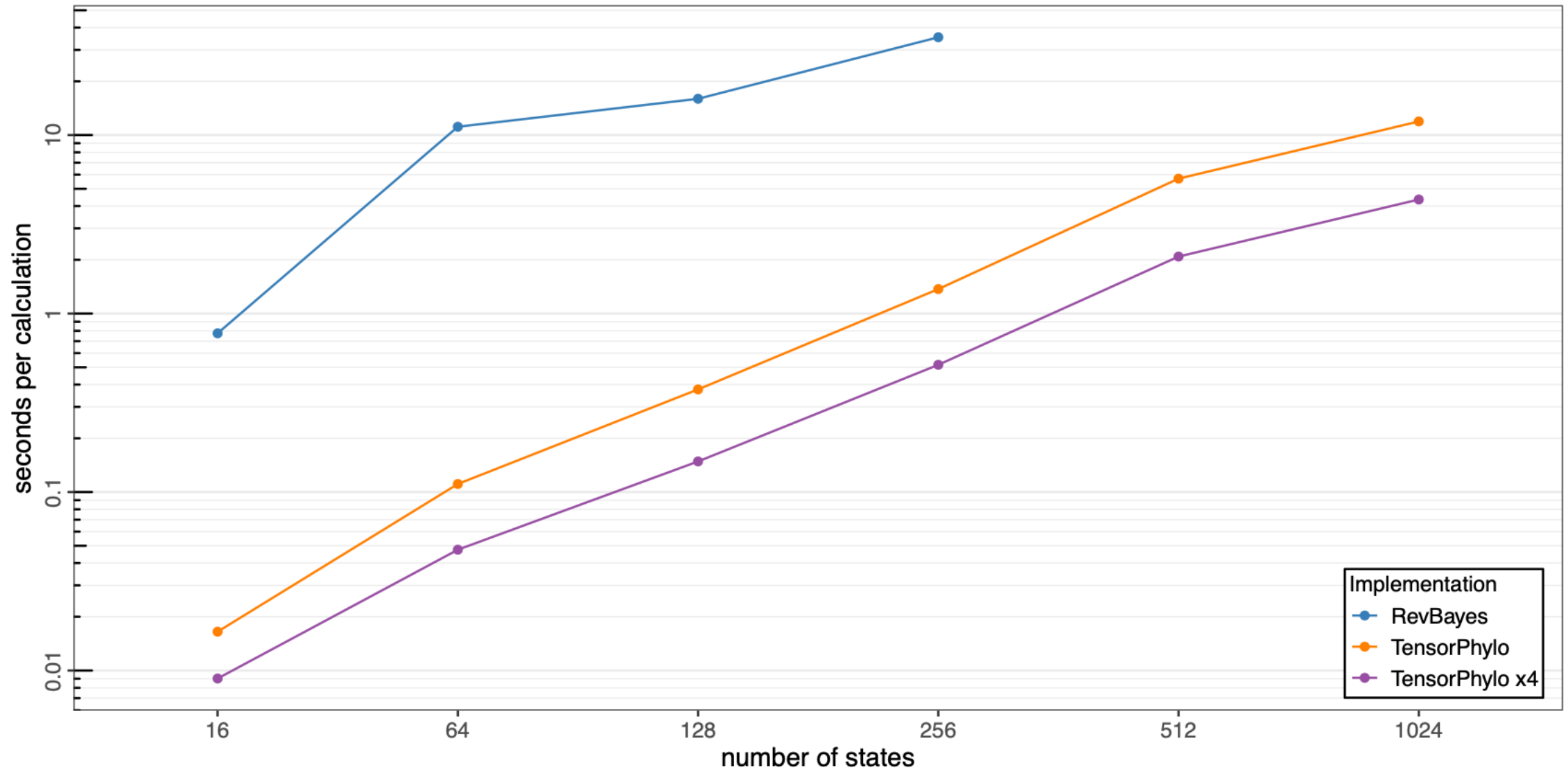
Performance: multiple states, no cladogenesis



Performance: two states, many tips



Performance: multiple states, with cladogenesis



6) Direcciones futuras en Biogeo

Feature Informed GeoSSE

